

Grade Calculator



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1. **Project Title:** Grade Calculator

2. **Team member's Name and ID:** I) MD. Siam Sheikh (2023100000033) & II) Tonni Biswas (2023100000016)

3. **Project Objective:**

- a. **Grade Calculation and Management:** Create a user-friendly interface that allows users to input and manage grades for multiple courses, including assignments, quizzes, and exams.
- b. **Accuracy:** Ensure accurate calculation of grades based on user-entered scores and weightage for each assessment component, taking into account different grading scales and methods.
- c. **Usability:** Design an intuitive and easy-to-use GUI that caters to users of varying technical backgrounds, making it accessible and straightforward to navigate.
- d. **Efficiency:** Provide an efficient means of calculating overall grades, saving users time and effort in manual grade calculations.
- e. **Grade Prediction:** Offer a feature that helps users predict their final grades based on their current performance and the weightage of remaining assessments.

4. **Stakeholder:**

- a. **Students:** Students are the primary users of the grade calculator. They use it to calculate and manage their grades for various courses and assessments.
- b. **Educators/Teachers:** Teachers may use the grade calculator to calculate and manage grades for their students, especially in cases where they need to weight different assessment components.
- c. **Academic Institutions:** Schools, colleges, and universities are stakeholders as they may encourage or require students and teachers to use a grade calculator to streamline the grading process.
- d. **Administrators:** Administrators within educational institutions may oversee the implementation and usage of the grade calculator to ensure its effectiveness and compliance with institutional policies.
- e. **Parents/Guardians:** Parents and guardians of students may be interested in monitoring their child's academic progress and could use the grade calculator to do so.

- f. **Developers/IT Department:** The team responsible for developing and maintaining the grade calculator application are stakeholders, as they need to ensure the software is functioning correctly and securely.

5. **Features of the projects [Functional requirement]:**

a. **User Interface:**

- Create an intuitive GUI that includes labels, input fields, buttons, and other elements for user interaction.
- Design a clean and user-friendly layout for easy navigation.

b. **Grade Entry:**

- Provide input field for users to enter their scores for various assessments within each course.

c. **Grade Calculation:**

- Implement logic to automatically calculate the overall grade for each course based on the entered scores.

d. **Grade Display:**

- Display the calculated grades for each course in a clear and organized manner within the GUI.
- Show the overall GPA (Grade Point Average) if applicable.

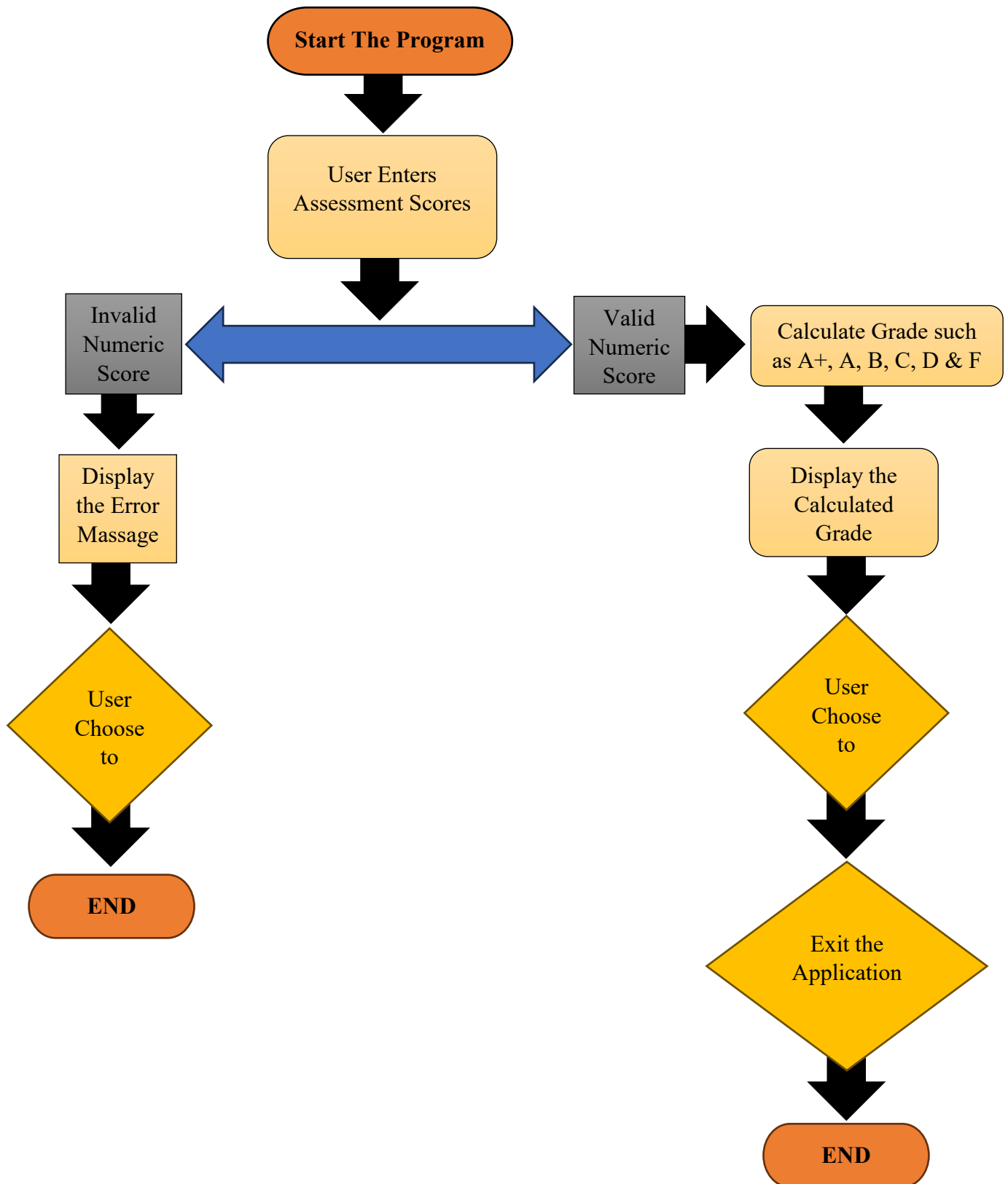
e. **Grade Scale Configuration:**

- Allow users to configure the grading scale used by the institution such as A⁺, A, B, C, D, F.

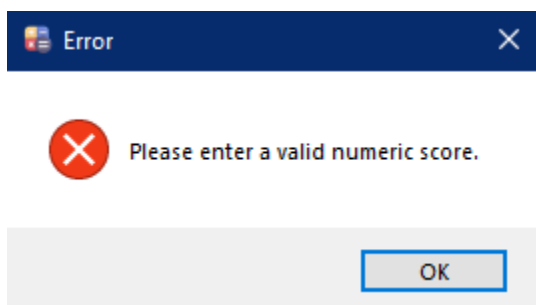
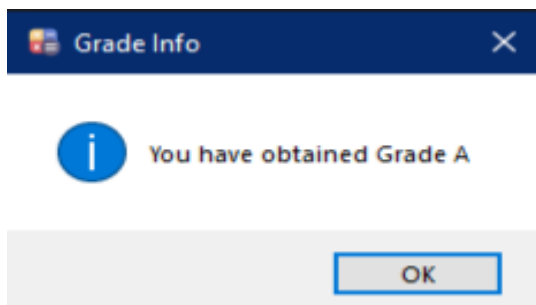
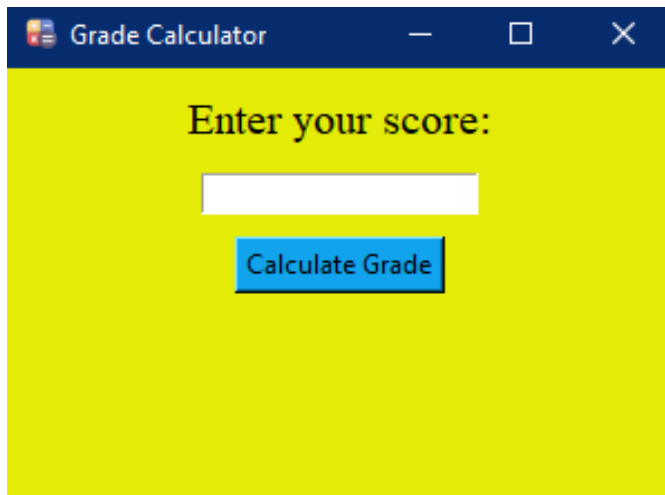
f. **Grade Prediction:**

- Provide a button or feature to estimate the final grade based on the user's current performance and the remaining assessments.

6. Draw the activity/ flow chart diagram:



7. Design the GUI (Graphical user interface of your system, scratch it using paper or any other tools before development):



```
import tkinter as tk
from tkinter import messagebox

def calculate_grade():
    try:
        score = float(entry_score.get())
        if score <= 100 and score >= 90:
            grade = "A+"
        elif score <= 89 and score >= 80:
            grade = "A"
        elif score <= 79 and score >= 70:
            grade = "B"
        elif score <= 69 and score >= 60:
            grade = "C"
        elif score <= 59 and score >= 40:
            grade = "D"
        else:
            grade = "F"
        messagebox.showinfo('Grade Info', f"You have obtained Grade {grade}")
    except ValueError:
        messagebox.showerror("Error", "Please enter a valid numeric score.")

app = tk.Tk()
app.title("Grade Calculator")
app.iconbitmap('F:/Programing/Python/calculator-3671700-3061901.ico')
app.config(bg='#e3eb07')
app.geometry('300x200')

label = tk.Label(app, text="Enter your score:", font=('times', 15), bg='#e3eb07')
label.pack(pady=10)

entry_score = tk.Entry(app)
entry_score.config(fg='#ed051c')
entry_score.pack()

button = tk.Button(app, text="Calculate Grade", command=calculate_grade)
button.config(bg='#11a4ed')
button.pack(pady=10)

app.mainloop()
```